Janada Jayathilaka

Associate Software Engineer | Full-Stack Developer | GenAI Engineer
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Portfolio: personal_portfolio

Personal Profile

Associate Software Engineer with experience in full-stack development, backend systems, and Generative AI applications. Skilled in building scalable RESTful APIs, AI-powered systems, and cloud-deployed applications using Node.js, FastAPI, and React. Experienced in RAG pipelines, multi-agent systems (LangChain/LangGraph), and CI/CD automation, with a strong focus on developing production-ready, high-performance software while continuously learning and adapting to emerging technologies.

Experience

Associate Software Engineer – Avanka IT (Pvt) Ltd **Feb 2026 – Present**

- Developing a cross-platform mobile application (React Native + Node.js) to streamline sales operations and improve workflow efficiency
- Building a local LLM-based document processing system enabling semantic search and fast querying of legal documents
- Designed and developed scalable RESTful APIs using Node.js and FastAPI
- Collaborated on full-stack development to deliver production ready features

Software Engineer Intern – Avanka IT (Pvt) Ltd **Sep 2025 – Feb 2026**

- Enhanced an existing online meeting platform using React and Node.js by implementing new features and improving system performance and user experience
- Designed and developed a centralized file repository system using React and Node.js, integrating SQL databases and AS400 for secure and scalable file storage and management
- Built a language translation backend using FastAPI (Python) and Hugging Face datasets, enabling real-time translation via RESTful APIs

Education

University of Kelaniya, Sri Lanka **Aug 2022 – Feb 2026**

- BSc (Hons) Electronics and Computer Science (GPA: 3.66)
- Relevant Coursework: Data Structures & Algorithms, OOP, Databases, Machine Learning

Taxila Central College, Horana

- G.C.E Advanced Level – Physical Science (Combined Mathematics -A, Chemistry – B, Physics- B, Z-score: 1.6)

Projects

AI Travel Agent Workflow (Backend Multi-Agent System) – (GitHub)

- Developed a backend multi-agent travel planning system using LangChain and LangGraph, implementing specialized agents for booking, flight search, and itinerary planning
- Designed agent workflows to coordinate between booker, flight, and planner agents for efficient task execution and response generation
- Tech Stack: Python, LangChain, LangGraph, LLMs

AI Q&A Multiagent System using RAG – (GitHub)

- Built a Retrieval-Augmented Generation (RAG) based Question Answering system to extract relevant information from documents and generate accurate, context-aware responses using LLMs.
- Designed an end-to-end pipeline including document preprocessing, text chunking, embedding generation, and semantic search, improving answer relevance and reducing hallucinations.
- Tech Stack: Python, LangChain, LangGraph, Pinecone, OpenAI

CI/CD Pipeline for Spring Boot Application – (GitHub)

- Built a Spring Boot application with a complete CI/CD pipeline using GitLab and Jenkins
- Containerized the application using Docker and configured Jenkins for continuous integration and deployment workflows
- Deployed the application on AWS with Nginx as a reverse proxy, integrated SonarQube for code quality analysis, and secured the system using Certbot (SSL)
- Tech Stack: Spring Boot, Java, GitLab CI/CD, Docker, Jenkins, Nginx, SonarQube, Certbot, MySQL, AWS

E-commerce (MERN Stack) Project – (GitHub)

- Developed a scalable e-commerce platform with authentication, admin dashboard, and payment integration.
- Implemented JWT-based authentication and developed an admin dashboard for managing products and orders, Images stored in AWS S3, and integrated the Stripe Payment Gateway.
- Designed CI/CD pipeline with GitHub Actions and deployed using Docker and AWS
- Tech Stack: React.js, Node.js, Express.js, MongoDB, AWS, Docker, CI/CD, GitHub Actions, Nginx, JWT

Spotify Clone(Backend Git Repository , Frontend Git Repository)

- Built a feature-rich music streaming platform with real-time playlist updates, user-artist interactions, and seamless audio playback.
- Implemented core functionalities such as secure login with JWT, song/artist search, song releases, and payment integration for premium access.
- Tech Stack: Spring Boot, React.js, CSS, MongoDB, JWT, Cloudinary

Skills

- **Programming Languages:**
  Python, JavaScript/ TypeScript, Java, SQL
- **Frameworks & Libraries:**
  Node.js, React.js, React Native, FastAPI, Spring Boot, Express.js, Tailwind CSS
- **Databases:** MySQL, MongoDB
- **Cloud & DevOps:**
  AWS, Docker, Jenkins, GitHub Actions, GitLab CI/CD, Nginx, Linux
- **AI & GenAI:**
  Large Language Models (LLMs), LangChain, LangGraph, RAG, Prompt Engineering, Vector Databases
- **Core Concepts:**
  RESTful APIs, Microservices Architecture, System Design, Data Structures & Algorithms

Leadership Experience and Volunteering

- **Electronics Student's Society – UOK** University of Kelaniya, Sri Lanka
  Junior Treasurer - Aug 2023 – Jul 2024
- **Electronics and Computer Science Club -UOK** University of Kelaniya, Sri Lanka
  Finance Lead - Jan 2023 – Sep 2023
- **UWU Robotic Death Race** University of Uva Wellassa, Sri Lanka
  Oct 2024

REFERENCES

Professor A.L.A.K. Ranaweera
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